Artificial Intelligence in Nephrology Education: A Multicenter Survey of Fellowship Trainees at Mayo Clinic

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Abstract

Artificial intelligence (AI) is increasingly recognized for its potential to enhance nephrology training and practice. However, the integration of AI into fellowship training remains inadequately explored. This study aimed to assess current AI utilization, perceptions, and educational needs among nephrology fellows at Mayo Clinic. A structured online survey was administered to 23 fellow–including those specializing in kidney transplantation and onco-nephrology–across three Mayo Clinic sites (Minnesota, Arizona, and Florida). The survey addressed domains such as current AI usage, perceived relevance of AI in clinical practice, interest in formal AI training, self-assessed comfort with AI integration, and barriers to adopting AI technologies in nephrology education. A total of 21 fellows (91% response rate) participated in the survey. 76% of respondents rated AI as moderately to highly relevant to nephrology. Similarly, 76% indicated a moderate to very high interest in receiving targeted AI training. Despite these favorable perceptions, 76% had rarely or never used AI in their clinical or research activities, and none reported any formal AI education. Interactive workshops emerged as the preferred modality for AI training (52%), with limited knowledge cited as the

primary barrier to adoption. Optimism was especially high regarding AI applications in predictive modeling (86%) and diagnostic imaging (81%), while confidence in AI for direct clinical decision-making remained cautious. There is significant interest among nephrology fellows in AI, along with a critical need for formal education and training. The enthusiasm for AI's potential contrasts with a cautious perspective towards its current use in clinical decision-making. Our study highlights the necessity for educational initiatives that bridge the knowledge gap and foster confidence in the appropriate use of AI technologies in Nephrology fellowship.